DGX Spark GPU Governor
GPU EDP and Bottleneck Analysis

Experiment 20260716_163946 only | 30 applications × 4 governors × 10 runs

GPU EDP = $avg\_gpu\_time\_sec^2 \times$ avg GPU power

CPU EDP omitted: CPU power is unavailable on DGX Spark.

CPU time is retained as application latency and bottleneck evidence.

Scope and metric

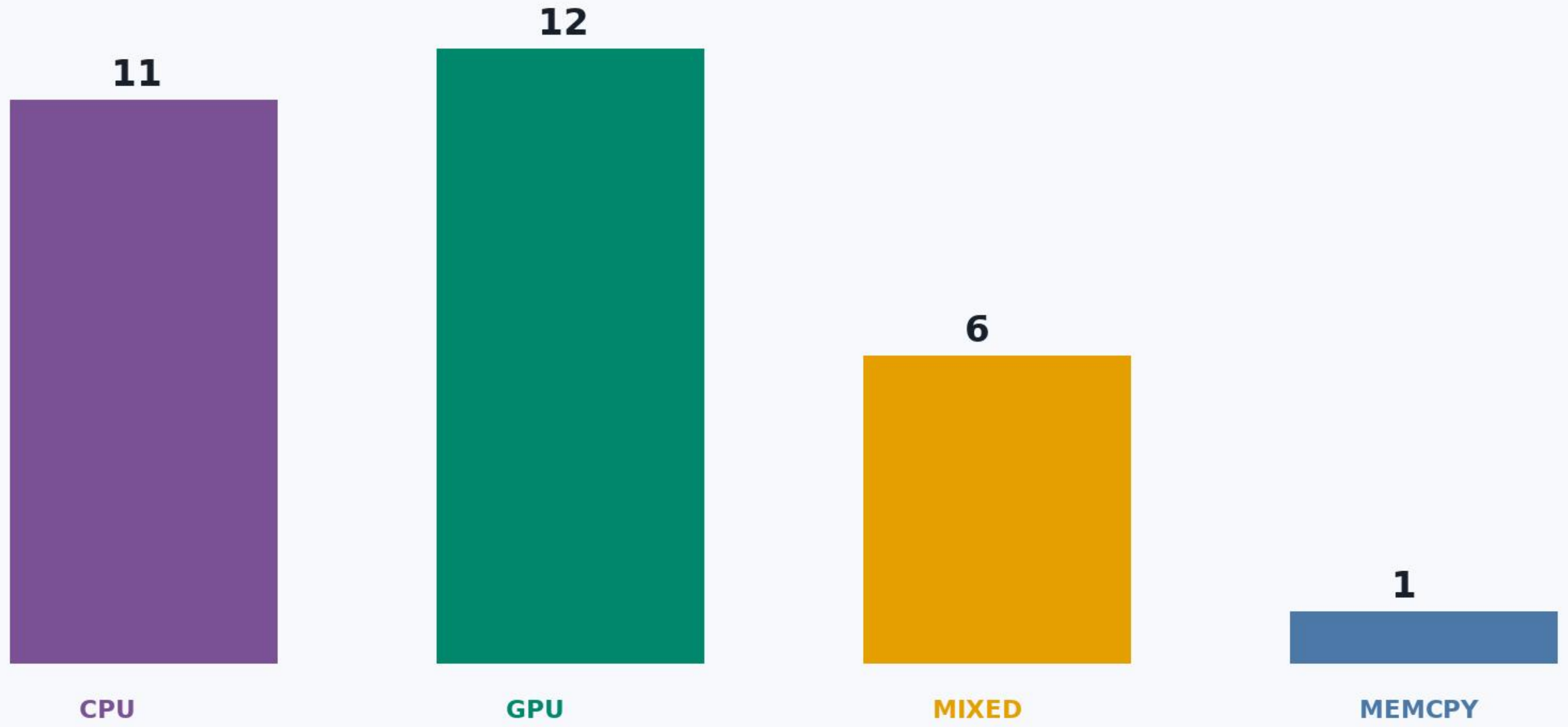
- Only 20260716_163946 is analyzed. The older 20260707_195535 experiment is excluded because it followed a different design and lacks CPU/GPU/memcpy time decomposition.
- Latency is avg_launch_time_sec = avg_cpu_time_sec + avg_gpu_time_sec + avg_gpu_memcpy_sec. The equality holds across all 1200 rows within CSV precision.
- GPU energy per launch = gpu_time × average GPU power. GPU EDP = GPU energy × gpu_time = gpu_time^2 × average GPU power.
- power_draw_w_avg equals the average of 10 Hz raw GPU power samples; the pair-wise difference is effectively zero.
- CPU EDP is not computed because Grace ARM does not expose CPU package power. Future x86 HPC analysis should use RAPL package energy and compute cpu_time^2 × cpu_power.

Headline findings

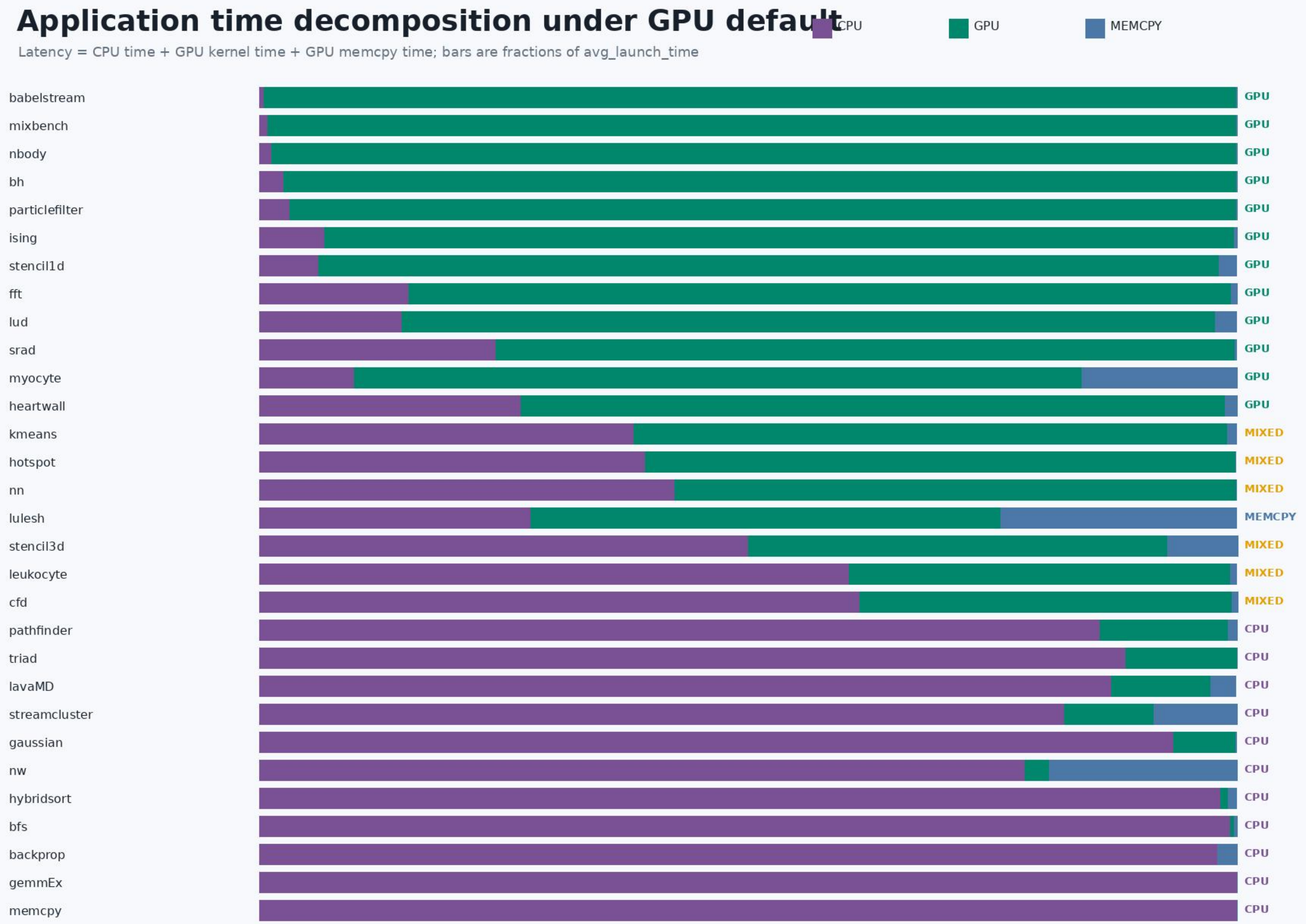
- Bottleneck classes: 11 CPU-bound, 12 GPU-bound, 6 mixed, and 1 memcpy-heavy application.
- Ondemand is not universally best under GPU EDP. Its median GPU EDP is approximately 4% worse than default, although it gives strong wins for backprop and nw.
- Powersave is generally harmful under GPU EDP because kernel slowdown is squared. Its median GPU EDP is several times default; backprop is the only valid <=5% latency win.
- Performance has no valid GPU EDP win and raises median GPU EDP by about 15%.
- Completed-launch filtering is mandatory: policies with fewer than five completed launches are failures, not zero-energy wins.
- SM activity is the most useful transparent workload signal. Low SM activity identifies candidates for an ondemand trial, but empirical validation is still required.

Bottleneck classification

Default-governor component shares; 70% threshold, memcpy threshold 20%



CPU time is analyzed as latency only. CPU EDP is omitted because CPU power is unavailable on DGX Spark.


Application time decomposition under GPU default
CPU
GPU
MEMCPY
Latency = CPU time + GPU kernel time + GPU memcpy time; bars are fractions of avg_launch_time
babelstream GPU
mixbench GPU
nbody GPU
bh GPU
particlefilter GPU
ising GPU
stencil1d GPU
fft GPU
lud GPU
srad GPU
myocyte GPU
heartwall GPU
kmeans MIXED
hotspot MIXED
nn MIXED
lulesh MEMCPY
stencil3d MIXED
leukocyte MIXED
cfd MIXED
pathfinder CPU
triad CPU
lavaMD CPU
streamcluster CPU
gaussian CPU
nw CPU
hybridsort CPU
bfs CPU
backprop CPU
gemmEx CPU
memcpy CPU

GPU EDP change by application and governor

GPU EDP = avg_gpu_time_sec² × avg GPU power; relative to GPU default

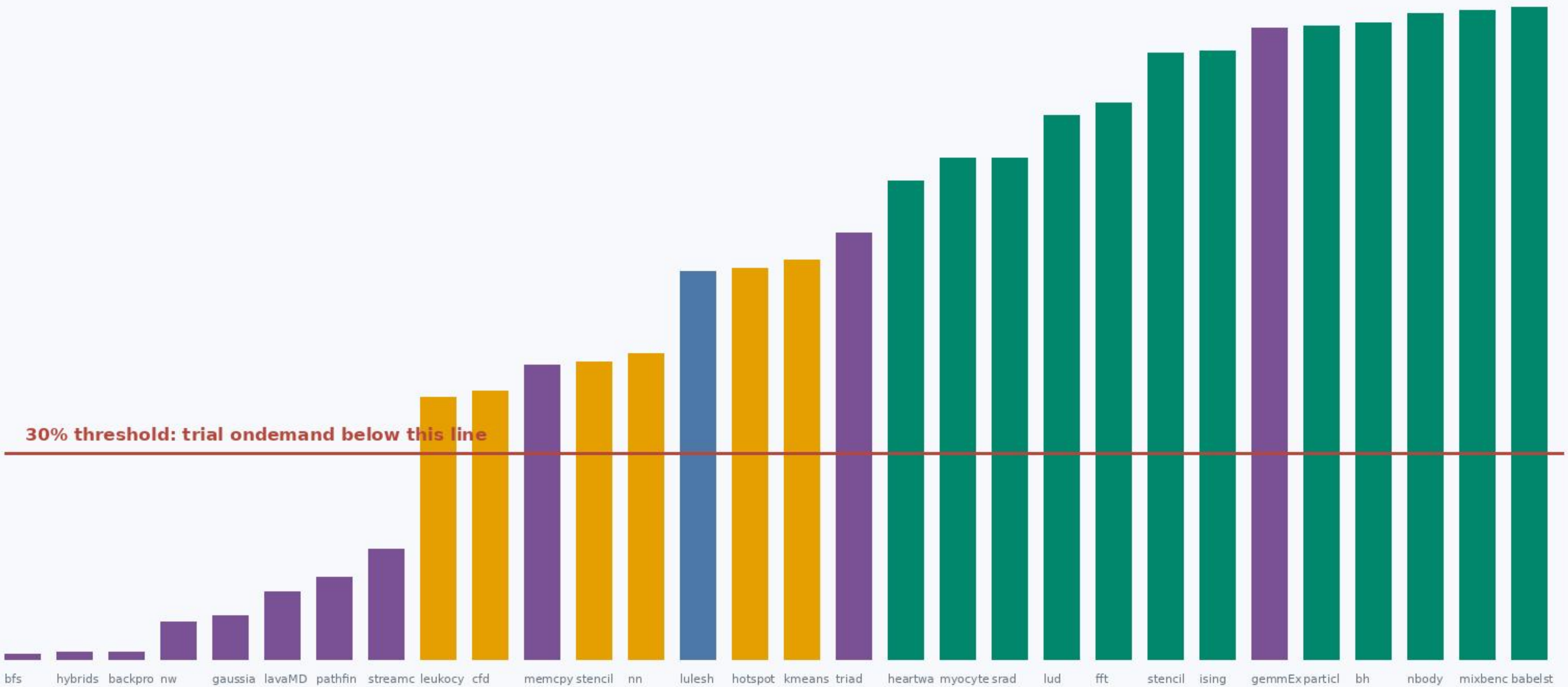
		ondemand	powersave	performance
babelstream	GPU	+23% | latency +0.7%	-46% | latency +26.0%	+19% | latency -0.2%
backprop	CPU	-52% | latency +1.9%	-50% | latency +2.5%	+54% | latency +0.2%
bfs	CPU	+297% | latency +1.5%	+308% | latency +0.7%	+113% | latency -0.3%
bh	GPU	-0% | latency +0.7%	+1426% | latency +852.7%	+3% | latency +0.6%
cfd	MIXED	+2% | latency +4.0%	+753% | latency +208.8%	+12% | latency +0.4%
fft	GPU	+1% | latency +1.9%	+2101% | latency +896.9%	+3% | latency +0.4%
gaussian	CPU	+69% | latency +2.0%	+846% | latency +18.9%	+90% | latency -0.8%
gemmEx	CPU	FAIL / N.A.	FAIL / N.A.	FAIL / N.A.
heartwall	GPU	-6% | latency +7.4%	+1731% | latency +742.0%	+4% | latency +0.2%
hotspot	MIXED	+1% | latency +3.4%	+1049% | latency +548.5%	+4% | latency +1.1%
hybridsort	CPU	+658% | latency +3.5%	+654% | latency +2.6%	+72% | latency +0.4%
ising	GPU	+5% | latency +0.3%	+244% | latency +361.4%	+8% | latency -0.5%
kmeans	MIXED	+20% | latency +25.8%	+1142% | latency +554.9%	+9% | latency +0.8%
lavaMD	CPU	-3% | latency -0.2%	+5155% | latency +104.8%	+59% | latency -0.3%
leukocyte	MIXED	+132% | latency +45.1%	+3080% | latency +418.7%	+5% | latency -0.2%
lud	GPU	+17% | latency +1.0%	+13% | latency +88.1%	+20% | latency +0.3%
lulesh	MEMCPY	+15% | latency +17.6%	+1313% | latency +321.3%	+16% | latency -0.0%
memcpy	CPU	FAIL / N.A.	FAIL / N.A.	FAIL / N.A.
mixbench	GPU	-3% | latency +0.3%	FAIL / N.A.	+4% | latency +0.7%
myocyte	GPU	+39% | latency +43.7%	FAIL / N.A.	+1% | latency +0.1%
nbody	GPU	-0% | latency +0.7%	+805% | latency +965.1%	+0% | latency +0.1%
nn	MIXED	-4% | latency +12.7%	+238% | latency +139.9%	+1% | latency +0.0%
nw	CPU	-41% | latency +0.0%	+804% | latency +8.5%	+53% | latency -0.4%
particlefilter	GPU	+15% | latency -0.3%	+1006% | latency +917.2%	+14% | latency -1.0%
pathfinder	CPU	+36% | latency +3.0%	+5031% | latency +133.1%	+66% | latency +0.2%
srad	GPU	+19% | latency +3.3%	+112% | latency +135.9%	+22% | latency +0.4%
stencil1d	GPU	+18% | latency +0.5%	+300% | latency +309.3%	+22% | latency +0.3%
stencil3d	MIXED	+0% | latency +3.2%	+768% | latency +181.1%	+29% | latency +0.3%
streamcluster	CPU	-14% | latency +21.8%	-16% | latency +16.8%	+47% | latency +0.7%
triad	CPU	-25% | latency +10.4%	+18% | latency +138.4%	+12% | latency +0.5%

CPU time fraction vs GPU EDP effect

Each point is a valid non-default policy; CPU time is latency, not CPU energy

SM activity predicts the bottleneck

Default-governor SM active percentage; proposed runtime threshold is 30%



Runtime rule validation

Rule: low SM activity triggers a guarded ondemand trial | exact match 90%, median EDP regret 0.0%

		predicted	empirical	EDP regret
babelstream	GPU	default	default	0.0%
backprop	CPU	ondemand	ondemand	0.0%
bfs	CPU	default	default	0.0%
bh	GPU	default	ondemand	0.4%
cfd	MIXED	default	default	0.0%
fft	GPU	default	default	0.0%
gaussian	CPU	default	default	0.0%
gemmEx	CPU	default	default	nan%
heartwall	GPU	default	default	0.0%
hotspot	MIXED	default	default	0.0%
hybridsort	CPU	default	default	0.0%
ising	GPU	default	default	0.0%
kmeans	MIXED	default	default	0.0%
lavaMD	CPU	ondemand	ondemand	0.0%
leukocyte	MIXED	default	default	0.0%
lud	GPU	default	default	0.0%
lulesh	MEMCPY	default	default	0.0%
memcpy	CPU	default	default	nan%
mixbench	GPU	default	ondemand	3.0%
myocyte	GPU	default	default	0.0%
nbody	GPU	default	ondemand	0.2%
nn	MIXED	default	default	0.0%
nw	CPU	ondemand	ondemand	0.0%
particlefilter	GPU	default	default	0.0%
pathfinder	CPU	default	default	0.0%
srad	GPU	default	default	0.0%
stencil1d	GPU	default	default	0.0%
stencil3d	MIXED	default	default	0.0%
streamcluster	CPU	default	default	0.0%
triad	CPU	default	default	0.0%

Runtime model and portability

Validated baseline rule

- Observe the application under GPU default. If sm_active_pct_avg < 30%, trial ondemand; otherwise retain default.
- Accept ondemand only when completed work remains valid, measured GPU EDP improves, and total launch latency stays within 5% of default. Otherwise revert immediately.

Portable features

- Use SM activity percentage, normalized SM clock, normalized video/memory clock, and GPU temperature. These describe workload pressure and governor state without encoding the GPU model.
- Do not use dram_active_pct from this experiment: it is zero in all 1200 rows. Do not use PCIe generation, DRAM type, or interconnect type as workload features.

Scale-up to x86 HPC

- Add CPU package energy from RAPL, compute CPU EDP independently, and then optimize a joint objective while retaining separate GPU and CPU latency budgets.